

Tensor computations

Border-rank deficiency forcing strict submultiplicativity at the tensor square

Difficulty: extreme **Importance:** interesting to the community

Status: [Lean verified](#)

Rating rationale: Forcing a saving at exactly two copies for every projective variety goes beyond the known eventual-power mechanism and presents a general structural barrier. The relationship between degeneration and repeated decomposition matters to the tensor-complexity community.

Negative resolution recorded 2026-09-11. Matthew J. Colbrook’s [complete manuscript](#), [Theorem 1 and Section 4 \(LaTeX source\)](#) constructs a smooth, irreducible, reduced, nondegenerate complex projective curve $X \subset \mathbb{P}^{11}$ and a point p with $\underline{R}_X(p) = 2$, $R_X(p) = 3$, and $R_{X \times X}(p \otimes p) = 9$. This refutes the exact universal implication below, using the specified Segre product with unrestricted complex coefficients. More generally, the construction delays strict submultiplicativity for any prescribed finite number of powers; the variety depends on that prescribed delay. It does not refute versions restricted to Segre or Veronese varieties or the known eventual-power saving. The full original source passed an [independent mathematical agent review](#). [Submission and verification record](#). That 2026-09-11 review is documented AI-assisted verification, not external human peer review; no priority claim is made. The original target and former difficulty and importance ratings are retained below as historical context.

Author feedback — 2026-09-17. In personal correspondence with Alex Townsend, Alessandra Bernardi, a coauthor of Conjecture 1.1, said she thought the argument works and agreed that it contradicts the conjecture in its stated generality. **The restrictions to Segre and Veronese varieties remain open.** This is informal mathematical feedback, not a formal referee report. [Correspondence summary](#).

Lean proof and verification evidence - 2026-09-22

The complete original conjecture is Lean verified, with a negative answer. The [formal proof at revision 775e8b1](#) constructs a reduced irreducible nondegenerate complex projective curve and a point with rank 3, border rank at most 2, and ordinary Segre tensor-square rank 9. These inequalities refute the full implication below. Decompositions allow arbitrary complex coefficients, repeated points, zero summands, the point at infinity, and independently chosen left and right tensor factors. The proof covers the whole variety, using an algebraic proof that the parameter cone equals the entire closed zero locus.

Lean formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, with substantial AI-agent assistance. **Matthew J. Colbrook** retains attribution for the original counterexample and mathematical proof. The conjecture authorship, manuscript, original statement and author feedback above remain preserved.

The [25 checked declarations](#) establish the projective, Zariski-closure and Segre semantics, the whole-image geometry, the rank and border-rank bounds, and the unconditional final theorems *NLA.TR27.projective_counterexample* and *NLA.TR27.original_conjecture_false*. The [statement boundary](#) was independently reviewed before implementation and remains byte-identical to its frozen receipt. Two fresh nonauthor agents approved the complete mathematical source and separately checked the actual verification evidence: [fidelity source review](#) and [completion review](#), and [correctness source review](#) and [completion review](#). These are independent AI reviews adapting the repository's Tau Ceti protocol, not external human peer review or official Tau Ceti endorsement.

The [successful non-root Linux run and operational audit](#) bind the immutable proof inputs to all 25 successful Comparator comparisons, default-kernel replay, authentic LeanCert kernel assertions, dependency revisions and isolation/rejection controls. Every selected declaration uses only *propext*, *Classical.choice* and *Quot.sound*. The pins are Lean 4.33.1, LeanCert 621a43d7cf21f87872392a01e874f2f1dbddc926 and Mathlib 0df444a360eaa60ab8c11dca51a86af692955474; the [shared source lock](#) pins the Comparator and control tools. This exact algebraic proof needs no numerical interval certificate.

The manuscript's additional smoothness, exact border rank two and arbitrary prescribed finite-delay results retain their manuscript and informal-review scope. They are unnecessary for this complete negative answer and are not claimed as Lean verified. The Segre- and Veronese-restricted questions remain open as recorded above. See the [project guide](#) and [formalization metadata](#). On the documented [non-root Linux environment](#), reproduce from the repository root with:

```
tools/lean/bootstrap.sh /absolute/path/to/verification-tools
tools/lean/verify.sh \
  tensor-computations/TR-27/lean \
  /absolute/path/to/verification-tools
```

Problem statement

Let V be a finite-dimensional complex vector space and let $X \subset \mathbb{P}(V)$ be an irreducible reduced nondegenerate complex projective variety. Nondegenerate means that X spans $\mathbb{P}(V)$. For $p \in \mathbb{P}(V)$, define $R_X(p)$ as the least number of points of X whose projective linear span contains p . Define $\underline{R}_X(p)$ as the least s for which p lies in the Zariski closure of $\{q : R_X(q) \leq s\}$.

Embed $X \times X$ in $\mathbb{P}(V \otimes V)$ by the Segre map $([x], [y]) \mapsto [x \otimes y]$. Is it always true that

$$\underline{R}_X(p) < R_X(p) \implies R_{X \times X}(p \otimes p) < R_X(p)^2?$$

Here $p \otimes p$ denotes the projective point represented by the tensor square of any representative of p . The product variety is precisely the Segre image just defined. For tensor-rank examples, this is the ordinary tensor product with both sets of modes retained; it does not silently merge corresponding modes into a Kronecker product.

Why it matters

An exact tensor decomposition can require more terms than arbitrarily close decompositions. The conjecture asks whether that deficiency already yields a strict saving when two copies are decomposed together. It connects ill-behaved rank limits with the cost of repeated multilinear computation.

References and status check

1. E. Ballico, A. Bernardi, F. Gesmundo, A. Oneto, and E. Ventura, *Geometric conditions for strict submultiplicativity of rank and border rank*, [arXiv:1909.03811v2](#), Conjecture 1.1; *Annali di Matematica Pura ed Applicata* **200** (2021), 187–210.
2. A. Oneto and E. Ventura, *Ranks of tensors: geometry and applications*, *Bollettino dell'Unione Matematica Italiana* **18** (2025), 751–778, Conjecture 4.9.

Status check — 2026-09-10

Checked the original v2, Conjecture 1.1 and Theorem 4.1, the 2025 restatement, and targeted strict-submultiplicativity searches through 2026. Theorem 4.1 proves the implication for all binary forms, a substantive subclass of the displayed target. The source also treats ternary cubics in Proposition 4.2 and its proof. A saving at some sufficiently large tensor power does not establish the required saving at the square. No general proof or counterexample was located; the product here retains both sets of factors.